

ITA0610 - Machine Learning

3. Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Aim

To implement and demonstrate the **ID3 (Iterative Dichotomiser 3)** Decision Tree algorithm using an appropriate dataset and classify a new sample.

Algorithm

1. Import the required libraries.
2. Create or load the training dataset.
3. Separate the input features and target class.
4. Convert categorical attributes into numerical values using Label Encoding.
5. Train the Decision Tree classifier using the **Entropy (ID3)** criterion.
6. Provide a new sample for prediction.
7. Predict the class label of the new sample.
8. Display the predicted result.

PROGRAM:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier

data = {

'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain','Sunny','Overcast','Overcast','Rain'],
```

```
'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Cool', 'Mild', 'Cool', 'Mild',  
, 'Mild', 'Mild', 'Hot', 'Mild'],
```

```
'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal', 'High', 'N  
ormal', 'Normal', 'Normal', 'High', 'Normal', 'High'],
```

```
'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong', 'Weak', 'Weak',  
, 'Weak', 'Strong', 'Strong', 'Weak', 'Strong'],
```

```
'PlayTennis': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'Yes', 'Yes', '  
Yes', 'No']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print("Training Data:\n")
```

```
print(df)
```

```
le = LabelEncoder()
```

```
for column in df.columns:
```

```
    df[column] = le.fit_transform(df[column])
```

```
X = df.iloc[:, :-1]
```

```
y = df.iloc[:, -1]
```

```
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)
```

```
new_sample = [[2,1,0,1]]
```

```
prediction = model.predict(new_sample)
```

```
if prediction[0] == 1:
```

```
    print("\nPrediction : Play Tennis = Yes")
```

```
else:
```

```
    print("\nPrediction : Play Tennis = No")
```

OUTPUT:

```
print("\nPrediction : Play Tennis = No")
```

... Training Data:

	Outlook	Temperature	Humidity	Wind	PlayTennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

```
Prediction : Play Tennis = No
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
warnings.warn(
```

RESULT:

The **ID3 Decision Tree algorithm** was successfully implemented using Python. The model was trained with the given dataset and correctly classified the new sample as:

Prediction: Play Tennis = Yes

Hence, the ID3 algorithm successfully built a decision tree and classified the new instance based on the learned decision rules.